Software Requirements Specification (SRS)

Airline Management System (AMS)

Team Members & Student Details

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1. Abstract

The **Airline Management System (AMS)** is a web-based, cloud-ready software application designed to streamline and automate core operations of commercial airline management. Grounded in Object-Oriented Programming (OOP) principles, the system delivers multi-role access control for Passengers, System Administrators, Pilots, Flight Hostesses, and Ground Staff. Passengers are empowered with real-time flight search, seat reservation, booking management, and ticket cancellation capabilities. Administrators maintain complete operational authority to schedule flights, track capacity, update flight statuses (On Time, Delayed, Cancelled, Reached Destination), and manage crew assignments while enforcing referential integrity. Built using PHP, MySQL, CSS3, and HTML5, and engineered for cloud deployment (via Vercel and cloud-hosted MySQL databases), the system provides an intuitive, robust, and scalable solution for modern aviation logistical challenges.

2. Technology Stack & Environment Requirements

2.1 Backend Architecture

- **Programming Language:** PHP 8.x (Object-Oriented Architecture with dynamic autoloading via spl_autoload_register())
- **Database Management System:** MySQL 8.0+ / MariaDB (Supports native connection pooling, prepared statements, and SSL/TLS encrypted database channels via MySQLi)

2.2 Frontend & UI Technologies

- **User Interface:** HTML5, Vanilla CSS3 (Custom responsive styling, variable tokens, dynamic micro-interactions)
- **Client Scripting:** JavaScript (ES6+) for interactive DOM updates and form validation

2.3 Hosting & Infrastructure

- **Deployment Platform:** Vercel (Serverless PHP Runtime integration via vercel.json)
- **Database Hosting:** Cloud MySQL instances (Aiven, Clever Cloud, PlanetScale, or local MySQL Workbench)
- **Environment Configuration:** 12-Factor App compliant environment variables (DB_HOST, DB_USER, DB_PASSWORD, DB_NAME, DB_PORT, DB_SSL_CA)

3. System Overview & User Roles

The system recognizes five distinct user roles with specific access levels:

1. **Passenger:** Authenticated customer capable of searching available flights, placing seat bookings, viewing ticket itineraries, and requesting cancellations.
2. **Admin:** System supervisor responsible for flight creation, status updating, staff onboarding/offboarding, and crew assignment.
3. **Pilot:** Flight operational crew member accessing personalized flight duty rosters.
4. **Flight Hostess:** In-cabin operational crew member accessing assigned flight schedules.
5. **Ground Staff:** Airport logistics personnel monitoring flight assignments.

4. Functional Requirements

4.1 User Authentication & Account Management

- **FR-1.1 User Registration:** The system shall allow prospective passengers to create an account by providing name, email address, and a password.
- **FR-1.2 Multi-Role Authentication:** The system shall authenticate users against their respective database role tables (Admin, Passenger, Pilot, Hostess, Ground Staff).
- **FR-1.3 Session Maintenance:** The system shall store session tokens and restrict access to role-specific dashboards based on session state.
- **FR-1.4 Logout:** The system shall allow users to securely terminate their session and clear stored session parameters.

4.2 Flight Operations & Searching (Passenger)

- **FR-2.1 Flight Search:** The system shall allow passengers to query available flights by specifying Source and Destination airports.
- **FR-2.2 Flight Availability Display:** The system shall output matching flight details including Flight Number, Origin, Destination, Departure Time, Arrival Time, Available Seats, and Status.

4.3 Booking & Reservation Management (Passenger)

- **FR-3.1 Ticket Booking:** The system shall enable authenticated passengers to book a seat on a selected flight provided available seats > 0.
- **FR-3.2 Automatic Seat & Code Generation:** The system shall automatically generate a unique alpha-numeric Booking Code and assign an available Seat Number upon reservation.
- **FR-3.3 Capacity Auto-Update:** The system shall decrement the available_seats count for a flight by 1 upon confirmation of a booking.
- **FR-3.4 Booking Itinerary View:** The system shall display a list of all active and past bookings for the logged-in passenger.
- **FR-3.5 Ticket Cancellation:** The system shall allow passengers to cancel active bookings on upcoming flights, freeing up the seat and incrementing available_seats by 1.

4.4 Flight Management (Admin)

- **FR-4.1 Flight Creation:** The system shall allow administrators to create new flights specifying Flight Number, Source, Destination, Departure Time, Arrival Time, and Total Seat Capacity.
- **FR-4.2 Status Updates:** The system shall enable admins to change flight statuses dynamically (On Time, Delayed, Cancelled, Reached Destination).
- **FR-4.3 Capacity Restrictions:** The system shall block new bookings if a flight status is set to Cancelled or Reached Destination.

4.5 Crew & Staff Scheduling (Admin & Staff)

- **FR-5.1 Staff Onboarding:** The system shall allow admins to register new Pilots, Flight Hostesses, and Airport Ground Staff.
- **FR-5.2 Crew Duty Assignment:** The system shall allow admins to map a Pilot, a Flight Hostess, and a Ground Staff member to a specific flight schedule.
- **FR-5.3 Roster Viewing:** The system shall allow Pilots, Hostesses, and Ground Staff to view their individual duty schedules.
- **FR-5.4 Deletion Safeguard:** The system shall prevent the deletion of a staff member if they are currently assigned to an active flight schedule.

5. Non-Functional Requirements

5.1 Performance Requirements

- **NFR-1.1 Response Time:** Database query execution and page render times shall remain below 1.5 seconds under standard connection conditions.
- **NFR-1.2 Concurrency:** The database connection wrapper (Database class) shall support multiple concurrent user connections without deadlock.

5.2 Security Requirements

- **NFR-2.1 Password Encryption:** Sensitive user passwords shall be hashed prior to database persistence.
- **NFR-2.2 SQL Injection Prevention:** All SQL queries interacting with user input shall utilize prepared statements or sanitized input bindings.
- **NFR-2.3 SSL/TLS Connectivity:** The system shall support secure, encrypted database connections via SSL certificates (MYSQLI_CLIENT_SSL).
- **NFR-2.4 Authorization & Access Control:** Unauthenticated users trying to access protected endpoints shall be automatically redirected to login.

5.3 Reliability & Availability

- **NFR-3.1 System Availability:** The system shall maintain 99.5% uptime when hosted on distributed cloud serverless infrastructure.
- **NFR-3.2 Referential Data Integrity:** Foreign key constraints (ON DELETE CASCADE / RESTRICT) shall strictly maintain relational database integrity.

5.4 Usability & Maintainability

- **NFR-4.1 Responsive Interface:** The web UI shall adapt fluidly across desktop, tablet, and mobile displays.
- **NFR-4.2 Modular OOP Architecture:** Business logic shall strictly reside inside modular PHP classes (User, Admin, Passenger, Pilot, Hostess, AirportStaff, Flight, Booking, Schedule).

6. Future Scope & Enhancement Possibilities

1. **Online Payment Gateway Integration:** Incorporate third-party payment gateways (e.g., Stripe, Razorpay, or PayPal) to process real-time ticket transactions and generate instant digital invoices.
2. **Interactive Seat Selection Map:** Enhance seat reservation from auto-generated seat numbers to a visual 2D aircraft seating matrix allowing passengers to select preferred window, aisle, or extra-legroom seats.
3. **Automated Notification Engine:** Implement automated SMS and Email services (via Twilio/SendGrid APIs) to broadcast flight delay alerts, gate changes, booking confirmations, and digital boarding passes.
4. **Dynamic Pricing & Baggage Allowance:** Introduce yield management algorithms to dynamically adjust ticket prices based on demand/seat occupancy and integrate baggage selection add-ons during booking.

5. **AI-Driven Customer Support Chatbot:** Integrate an AI virtual assistant to handle passenger queries regarding flight schedules, baggage policies, and booking status inquiries 24/7.